M2S-01: Step 1 — Discover: Understand SaaS Differences

> **Series:** M2S — Managed to SaaS Migration | **Notebook:** 1 of 9 | **Phase:** Plan | **Step:** Discover | **Created:** March 2026 | **Last Updated:** 04/26/2026

The first step in any Managed-to-SaaS migration is understanding what you are moving to and why. This notebook helps you document the benefits of Dynatrace SaaS for your organization, take inventory of your current Managed environment, and confirm your use cases and goals for the upgrade.

> **M2S Migration Journey — 3 Phases / 9 Steps**
>
> **Plan: 1. Discover** | 2. Strategize | 3. Design
>
> **Upgrade:** 4. Prepare | 5. Execute | 6. Integrate
>
> **Run:** 7. Expand | 8. Enable | 9. Optimize

Sprint 1.337 (April 2026) Updates Affecting M2S

Three sprint-1.337 changes affect a Managed → SaaS migration:

1. **OneAgent primary fields/tags at source** — Latest Dynatrace SaaS tenants gain primary fields (`dt.security_context`, `dt.cost.costcenter`, `dt.cost.product`) and customer-defined primary tags as top-level attributes. Consider this in the M2S-03 (Design) step when deciding what tag/security-context model to recreate post-migration.
2. **Configuration API → Settings v2 acceleration** — Dynatrace Managed exposes the older Configuration API for many resources; the SaaS target's Settings v2 surface now covers more of those endpoints. Migration tooling (M2S-04/05) should target Settings v2 wherever possible. Plan for the legacy/v2 split where Managed has a unique Configuration API endpoint.
3. **Platform tokens** for new automation. Classic `dt0c01` still works for legacy paths, but new SaaS pipelines should default to `dt0s16` / `dt0s01` Platform tokens with

`Authorization: Bearer …`.

Extensions 3rd-gen API (ToDo #1) — if Managed has custom 2nd-gen Extensions, migration to SaaS is also the natural moment to graduate to 3rd-gen via the Dynatrace API Application → Extensions surface.

Table of Contents

Inventory Categories

A complete discovery must cover ALL of these categories — missing any one will create surprises during execution:

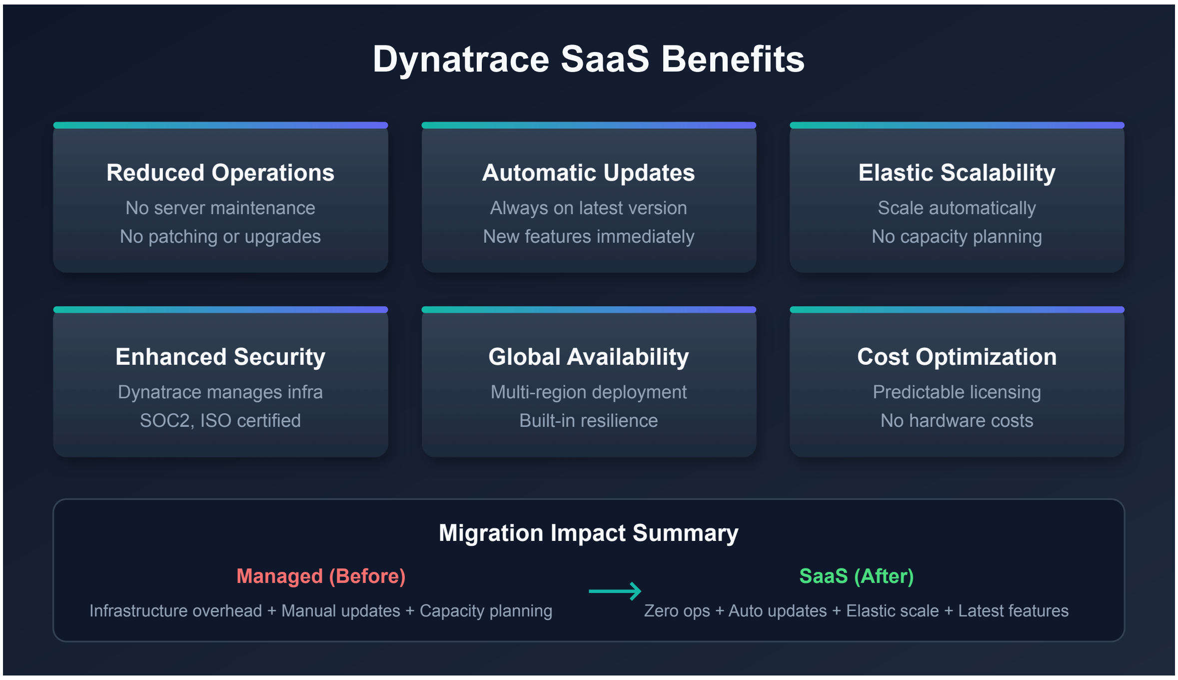
Category	What to Inventory	Migration Method
Configurations and settings	All environment-level settings, entity-level settings	Automated via SaaS Upgrade Assistant
Credential Vault	All stored credentials, certificates	Manual — secrets cannot be exported
API tokens	All tokens and their scopes	Manual — recreate with minimal scopes

Category	What to Inventory	Migration Method
Extensions	OneAgent extensions, ActiveGate extensions — are they current or need upgrade?	Manual — evaluate for Extensions 2.0
External and custom sources	Cloud integrations (AWS/Azure/GCP), Kubernetes integration, log ingest, custom metrics	Manual — reconfigure each source
Other integrations	ITSM, CMDB, reports, data lakes, CI/CD pipelines	Manual — update endpoints and tokens
Monitoring components	OneAgent instances (hosts, PaaS, K8s/OpenShift), ActiveGate instances (routing, extensions, synthetic, zRemote)	Automated (agent redirect)
Dashboards	All dashboards, their owners, and management zone filters	Automated via SaaS Upgrade Assistant

> **Tip:** Consider environment clean-up during discovery. Excessive or legacy configuration items can be left behind. Most items like tags or management zones migrate in full, but others like dashboards should be reviewed and migrated selectively.

Prerequisites

Requirement	Details
Dynatrace Managed	Active Managed environment with administrator access
Dynatrace SaaS Tenant	Provisioned SaaS environment (or planning to provision)
API Tokens	Tokens with `entities.read`, `settings.read` scopes on Managed
Stakeholder Access	Ability to document and share findings with decision-makers



1. Why Upgrade to SaaS

Moving from Dynatrace Managed to SaaS is not just a hosting change — it unlocks a fundamentally different platform architecture. Understanding these differences helps you build the business case and set expectations with stakeholders.

Infrastructure and Operations

Area	Managed	SaaS
Cluster management	Customer-managed servers, storage, networking	Fully managed by Dynatrace
Scaling	Manual capacity planning and provisioning	Automatic scaling with licensing
Updates	Manual cluster patching (scheduled downtime)	Automatic bi-weekly updates (zero downtime)
Availability	Customer-managed HA/DR	99.5%+ SLA with built-in redundancy
Security patches	Customer responsibility to apply	Automatic, managed by Dynatrace

Platform Capabilities

SaaS provides capabilities that are not available on Managed:

Capability	Description
Grail	Petabyte-scale data lakehouse — unified querying across logs, metrics, traces, events, and entities
DQL	Dynatrace Query Language — context-aware queries replacing USQL
Notebooks	Interactive, collaborative analysis documents (you are reading one now)
AppEngine	Build and deploy custom Dynatrace apps
AutomationEngine	Advanced workflow engine for alerting, remediation, and orchestration
Dynatrace Assist	AI-powered assistant for natural-language queries and analysis
OpenPipeline	Custom data processing, routing, and enrichment at ingest
Business Analytics	Full business event tracking and conversion funnel analysis

Operational Cost Reduction

Migrating to SaaS eliminates operational overhead associated with running Managed clusters:

- No hardware procurement, racking, or lifecycle management
- No cluster OS patching or Dynatrace server upgrades
- No capacity planning for Cassandra, Elasticsearch, or transaction storage
- No backup/restore procedures for cluster data
- No need for dedicated Managed cluster administrators

Enhanced Security and Compliance

Compliance	Coverage
SOC 2 Type II	Annual audit of security controls

Compliance	Coverage
ISO 27001	Information security management certification
HIPAA	Health data handling compliance
FedRAMP	US government security authorization (select regions)
Automatic patching	Security vulnerabilities addressed without customer action

2. Review Your Current Environment

Before planning the migration, take a complete inventory of what exists in your Managed environment. This inventory serves two purposes: sizing your SaaS license and identifying configurations that need migration.

> **Note:** DQL queries below run against your **SaaS tenant** (where entity data has been synced or where agents are already reporting). For Managed-only environments, use the **Entities API v2** or **Dynatrace UI** alternatives shown after each query.

Host Inventory

```
// Count total monitored hosts
fetch dt.entity.host
| summarize hostCount = count()

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | summarize hostCount = count()
```

```
// Hosts grouped by OS type
fetch dt.entity.host
| summarize hostCount = count(), by:{osType}
| sort hostCount desc

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | summarize hostCount = count(), by:{osType}
// | sort hostCount desc
```

Managed alternative (Entities API v2):

```
# Total host count
curl -s "https://{managed-url}/api/v2/entities?
entitySelector=type(HOST)&amp;pageSize=1" \
  -H "Authorization: Api-Token {TOKEN}" | jq '.totalCount'
```

Service Inventory

```
// Count total monitored services
fetch dt.entity.service
| summarize serviceCount = count()

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes SERVICE
// | summarize serviceCount = count()
```

```
// Services grouped by technology
fetch dt.entity.service
| summarize serviceCount = count(), by:{serviceType}
| sort serviceCount desc

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes SERVICE
// | summarize serviceCount = count(), by:{serviceType}
// | sort serviceCount desc
```

Application Inventory

```
// Count monitored web applications
fetch dt.entity.application
| summarize appCount = count()

// Note: smartscapeNodes WEB_APPLICATION is not yet available on Grail
// Continue using fetch dt.entity.application until Smartscape coverage
expands
```

Synthetic Test Inventory

```
// Count synthetic monitors
fetch dt.entity.synthetic_test
| summarize syntheticCount = count()

// Note: smartscapeNodes SYNTHETIC_TEST is not yet available on Grail
// Continue using fetch dt.entity.synthetic_test until Smartscape coverage
expands
```

```
// Synthetic tests by type
fetch dt.entity.synthetic_test
| summarize testCount = count(), by:{type}
| sort testCount desc

// Note: smartscapeNodes SYNTHETIC_TEST is not yet available on Grail
// Continue using fetch dt.entity.synthetic_test until Smartscape coverage
expands
```

OneAgent Version Assessment

Understanding your OneAgent version spread is important because SaaS supports a rolling 9-month (Standard) / 12-month (Enterprise) version window. Agents older than this window must be updated before or during migration.

```
// OneAgent versions across hosts
fetch dt.entity.host
| summarize hostCount = count(), by:{installerVersion}
| sort hostCount desc

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | summarize hostCount = count(), by:{installerVersion}
// | sort hostCount desc
```

ActiveGate Assessment

```
// ActiveGate inventory
fetch dt.entity.active_gate
| fieldsAdd entity.name
| summarize activeGateCount = count()

// Note: smartscapeNodes ACTIVE_GATE is not yet available on Grail
// Continue using fetch dt.entity.active_gate until Smartscape coverage
expands
```

Environment Size Summary

Record your findings in this table:

Entity Type	Count	Notes
Hosts	___	Include OS breakdown
Services	___	Include technology breakdown
Applications	___	Web and mobile
Synthetic Tests	___	HTTP, browser, scripted
ActiveGates	___	Environment and cluster
Management Zones	___	Will migrate as-is
Dashboards	___	Classic dashboards migrate; rebuild as modern recommended

> **Important:** If your host count exceeds 25,000, you will need multiple SaaS tenants. Discuss tenant topology with your Dynatrace account team before proceeding.

3. Confirm Use Cases and Goals

Every migration needs clearly documented motivations and success criteria. Use the table below to identify which benefits matter most to your organization and align them with concrete goals.

Common Migration Motivations

Motivation	SaaS Benefit	Success Metric
Eliminate cluster maintenance	Fully managed infrastructure	Zero hours spent on cluster patching
Access new capabilities	Grail, Notebooks, AppEngine, Dynatrace Assist	Teams actively using SaaS-exclusive features
Improve availability	99.5%+ SLA	Reduced monitoring downtime incidents
Reduce operational cost	No hardware, patching, capacity planning	Measured reduction in Managed infrastructure spend
Enhance security posture	Managed infrastructure, automatic security patches	Faster time-to-patch for vulnerabilities
Simplify compliance	SOC 2 Type II, ISO 27001, HIPAA	Inherited compliance controls
Consolidate tenants	Single SaaS platform	Fewer environments to manage
Enable self-service	IAM policies, Grail permissions	Teams can query their own data without admin requests

Documenting Your Goals

For each selected motivation, document:

1. **Current pain point** — What problem does this solve today?
2. **Expected outcome** — What does success look like after migration?
3. **Measurement** — How will you verify the goal was achieved?
4. **Timeline** — When should this benefit be realized (during migration or post-migration)?

> **Tip:** Share this goals document with stakeholders before proceeding to Step 2 (Strategize). Alignment on goals prevents scope creep and sets realistic expectations.

4. SaaS Upgrade Assistant Overview

The **SaaS Upgrade Assistant** is Dynatrace's primary migration tool for Managed-to-SaaS upgrades. Understanding its capabilities early helps you plan what will be automated versus what requires manual effort.

How It Works

1. **Export** — Download configuration archive from your Managed Cluster Management Console
2. **Upload** — Import the archive into the SaaS Upgrade Assistant app on your target SaaS tenant
3. **Review** — Configurations are grouped by type with error highlighting for incompatible items
4. **Edit** — Fix failed or incompatible configurations individually or in bulk
5. **Deploy** — Push selected configurations to your SaaS environment
6. **Track** — Monitor progress and download deployment result reports (CSV)

Key Features

Feature	Description
Automated configuration import	Most settings and dashboards migrate automatically

Feature	Description
Selective import	Smart dependency-aware import — choose what to deploy per wave
Dashboard ownership migration	Updates dashboard owners from Managed to SaaS user identifiers
Entity ID adjustment	Handles entity ID changes between environments
Bulk editing	Edit and correct failed configurations in batch
Preview changes	Review all changes before deploying
Progress tracking	Real-time status with CSV export of results

Requirements

Requirement	Details
Managed version	1.294 or later
Version alignment	Best results when Managed and SaaS are on the same major version (e.g., both 1.294.x)
IAM policy	`upgrade-assistant:environments:write` assigned in Account Management
Installation	Install the SaaS Upgrade Assistant app on your target SaaS tenant

> **Tip:** Contact your Dynatrace account team for guided migration support. They can help plan waves and troubleshoot incompatible configurations.

5. What Migrates and What Doesn't

Understanding portability constraints upfront prevents surprises during execution. The SaaS Upgrade Assistant handles most configuration types, but some items require manual recreation.

Portable via SaaS Upgrade Assistant

Configuration Type	Notes
Settings and configurations	Application detection, service detection, data privacy
Management zones	Migrate as-is; consider converting to Grail permissions post-migration
Auto-tagging rules	Tag definitions and conditions
Alerting profiles	Alert filtering and notification routing
Classic dashboards	Migrate automatically; rebuild as modern dashboards recommended
Service detection rules	Custom service naming and merging
Request attributes	Capture rules for request metadata
Calculated metrics	Service, log, and RUM calculated metrics
Maintenance windows	Scheduled suppression windows

Requires Manual Recreation

Configuration Type	Why	Recommended Action
Credentials Vault entries	Security — secrets cannot be exported	Re-create in SaaS Credentials Vault
API tokens	Security — tokens are environment-specific	Generate new tokens in SaaS
Webhook endpoints	URLs may differ for SaaS network paths	Reconfigure notification integrations
Synthetic private locations	ActiveGate-bound, environment-specific	Deploy new ActiveGates, recreate locations
Custom extensions (1.0)	EF1 deprecated — must rebuild as Extensions 2.0	Rebuild using Extensions 2.0 framework
Network zones	Environment-specific network configuration	Recreate in SaaS if needed

Configuration Type	Why	Recommended Action
Plugin-based integrations	Legacy plugin framework	Migrate to Extensions 2.0 or ActiveGate extensions

Non-Portable (Data)

> **Important:** Historic data does **not** migrate. This includes metrics, logs, traces, user sessions, problems, and detected events. Plan for a baseline period where both Managed and SaaS run in parallel so SaaS accumulates its own history.

Data Type	Retention Impact
Metrics	New baseline starts from SaaS go-live
Logs	No historic log data in SaaS
Traces/spans	Distributed traces start fresh
User sessions	RUM session data not transferred
Problems	detected problem history stays in Managed
Deployment events	Release tracking starts fresh

6. Step Completion Checklist

Before proceeding to Step 2, confirm that you have completed each item:

Checkpoint	Status
SaaS benefits documented and shared with stakeholders	[ ]
Current environment inventory complete (hosts, services, applications, synthetics)	[ ]
OneAgent version spread assessed (any agents outside support window identified)	[ ]
ActiveGate inventory documented	[ ]

Checkpoint	Status
Migration use cases and goals documented with success metrics	[ ]
SaaS Upgrade Assistant requirements understood (version, IAM, installation)	[ ]
Non-portable items identified and manual recreation plan noted	[ ]
Historic data limitation communicated to stakeholders	[ ]

Next Step

> **M2S-02: Step 2 — Strategize** — Define your migration approach, timeline, and risk assessment. Choose between big-bang and phased migration, plan your parallel-run period, and identify dependencies and risks.

Additional Resources

- SaaS Upgrade Assistant Documentation
- SaaS Upgrade Assistant on Dynatrace Hub
- Upgrading from Dynatrace Managed to SaaS
- Grail Data Lakehouse
- DQL Reference

Summary

In Step 1, you:

- Documented the key benefits of Dynatrace SaaS compared to Managed
- Inventoried your current environment (hosts, services, applications, synthetics, ActiveGates)
- Confirmed your migration use cases and goals with measurable success criteria
- Understood the SaaS Upgrade Assistant as the primary migration tool

- Identified what migrates automatically, what requires manual recreation, and what data does not transfer

---